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Digital Signal Processing Miniproject Report

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Project Title:

Voice Controlled DC Motor Speed Regulation

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Project Report

Title: Voice-Controlled DC Motor Speed Regulation

1.Abstract

This project presents the design and implementation of a voice-controlled DC motor speed regulation system using digital signal processing (DSP) techniques and embedded hardware. Traditional motor control relies on manual switches, which may be inconvenient or inaccessible in certain applications. By employing voice commands, the system enables intuitive, hands-free operation. The methodology integrates MATLAB for signal processing and feature extraction with an ESP32 microcontroller for real-time motor control. Recorded voice signals are filtered using FIR/IIR filters to reduce noise, and key features such as energy and pitch are extracted for command classification. Commands such as *SLOW* and *FAST* are transmitted to the ESP32 via serial communication, where PWM signals generated through an L298N motor driver regulate motor speed. Experimental results validate the system's accuracy in recognizing commands and adjusting motor speed accordingly, with real-time feedback displayed on an LCD. The proposed framework demonstrates a cost-effective, efficient, and scalable solution for voice-interactive motor control, with potential applications in robotics, automation, and assistive technologies.

2. Introduction

Controlling machines and robots using natural human interaction methods such as voice commands has gained considerable attention across research and industrial domains. Traditional motor control typically uses manual switches or control panels, which can be limited in convenience and accessibility, especially for people with disabilities. Voice-controlled systems offer an intuitive, hands-free alternative that enhances interaction quality. While many existing systems rely on computational heavy machine learning techniques, this project leverages the efficiency of digital signal processing methods like FIR/IIR filters within MATLAB for real-time voice command recognition and motor speed regulation. The integration of MATLAB with embedded microcontrollers such as Arduino and ESP32 provides a practical hardware-software platform that is cost-effective and responsive.

3.Literature Review:

Voice-controlled motor systems have emerged as an intuitive solution for human-machine interaction, offering hands-free control and improved accessibility.

Andriyan and Aji (2021) demonstrated DC motor speed regulation using speech recognition and digital signal processing techniques, establishing foundational methods for voice-based motor control. Omweno et al. (2023) enhanced this approach by integrating voice and gesture control for DC motor fans, enabling control of both speed and direction while improving system accessibility for diverse users. Sivaranjani and Prakash (2025) extended voice control applications to BLDC motors using modular multilevel inverters, demonstrating the scalability of voice commands in industrial motor control systems. Khalid et al. (2024) developed Arduino-based trainer kits for DC motor speed control through voice commands, highlighting the educational value and cost-effectiveness of such implementations.

Despite these advancements, most existing systems depend on computationally intensive machine learning algorithms requiring significant processing power. This project addresses this limitation by employing efficient digital signal processing methods, specifically FIR/IIR filtering combined with energy and pitch-based feature extraction. This approach provides a lightweight, real-time solution suitable for resource-constrained embedded systems like ESP32, while maintaining high accuracy and responsiveness in command recognition and motor control.

4.Objectives

1. To record and process voice commands in MATLAB for real-time recognition using a sampling frequency of 16 kHz.
2. To apply FIR low-pass filtering technique for speech preprocessing and noise reduction from recorded audio signals.
3. To extract meaningful speech features including signal energy and pitch frequency using FFT analysis for command classification.
4. To classify the processed commands into motor control actions (SLOW and FAST) based on energy and pitch thresholds, and transmit them to ESP32 via serial communication.
5. To implement motor speed regulation using PWM signals on ESP32 with L298N motor driver and display real-time motor status on LCD.

5.Introduction to FIR Filter

A Finite Impulse Response (FIR) filter is a digital filter whose impulse response settles to zero in a finite duration. Its output depends only on the current and a finite number of past input samples, given by:

$$y[n] = \sum_{k=0}^{N} b_k \, x[n-k]$$

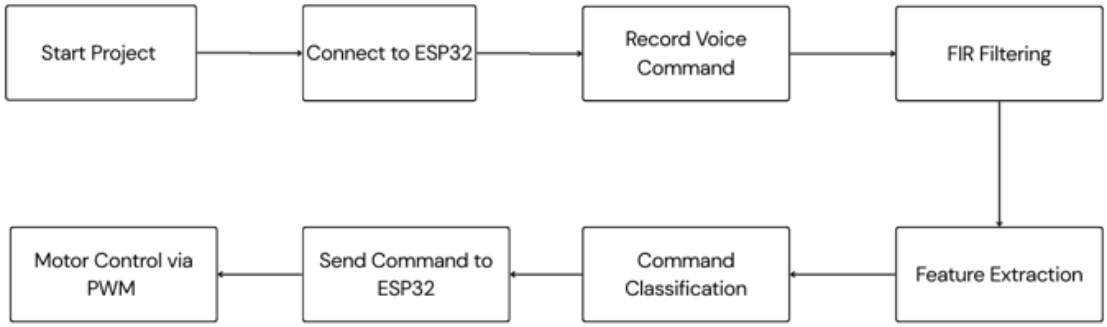
FIR filters are always stable, can be designed for linear phase, and are widely used in audio, communication, and image processing applications.

Characteristics of FIR Filters:

1. **Finite response:** The output settles to zero after a finite number of samples.
2. **Always stable:** No feedback is used, so the filter cannot become unstable.
3. **Linear phase:** With symmetric or anti-symmetric coefficients, the filter can preserve the waveform of signals.
4. **Simple design methods:** FIR filters can be designed using windowing, frequency sampling, or optimization techniques.

6.BLOCK DIAGRAM

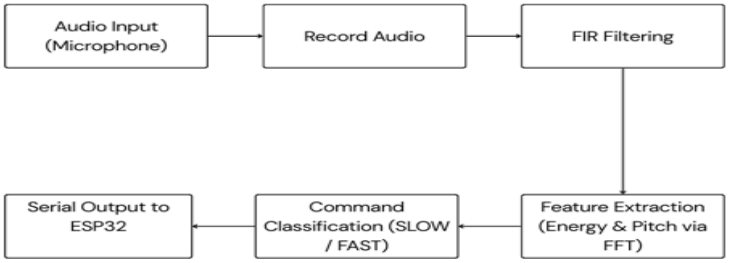
A] Full Workflow Diagram (Voice-Controlled Motor System)



Blocks and Functions:

1. **Start Project** – Initializes the system and prepares MATLAB/ESP32 for operation.
2. **Connect to ESP32** – Establishes serial communication between MATLAB and the ESP32.
3. **Record Voice Command** – Captures the user's voice through a microphone.
4. **FIR Filtering** – Applies a low-pass FIR filter to remove high-frequency noise from the audio.
5. **Feature Extraction (Energy & Pitch)** – Extracts key characteristics of the signal to identify the command.
6. **Command Classification (SLOW / FAST)** – Determines the motor command based on extracted features.
7. **Send Command to ESP32** – Transfers the classified command via serial communication.
8. **Motor Control via PWM** – ESP32 adjusts motor speed according to the command (SLOW or FAST).
9. **Display on LCD** – Updates the motor status on an LCD connected to ESP32.

B] Audio/Sensor Processing Diagram (Voice Data Focus)



Blocks and Functions:

1. **Audio Input (Microphone)** – Captures the raw voice signal from the user.
2. **Record Audio** – Stores the signal in MATLAB for processing.
3. **FIR Filtering (Low-pass)** – Removes unwanted high-frequency components.
4. **Feature Extraction (Energy & Pitch via FFT)** – Calculates signal energy and identifies dominant frequency (pitch).
5. **Command Classification (SLOW / FAST)** – Determines the command based on energy and pitch thresholds.
6. **Serial Output to ESP32** – Sends the processed command to ESP32 for motor control.

7. DESIGN AND WORKING

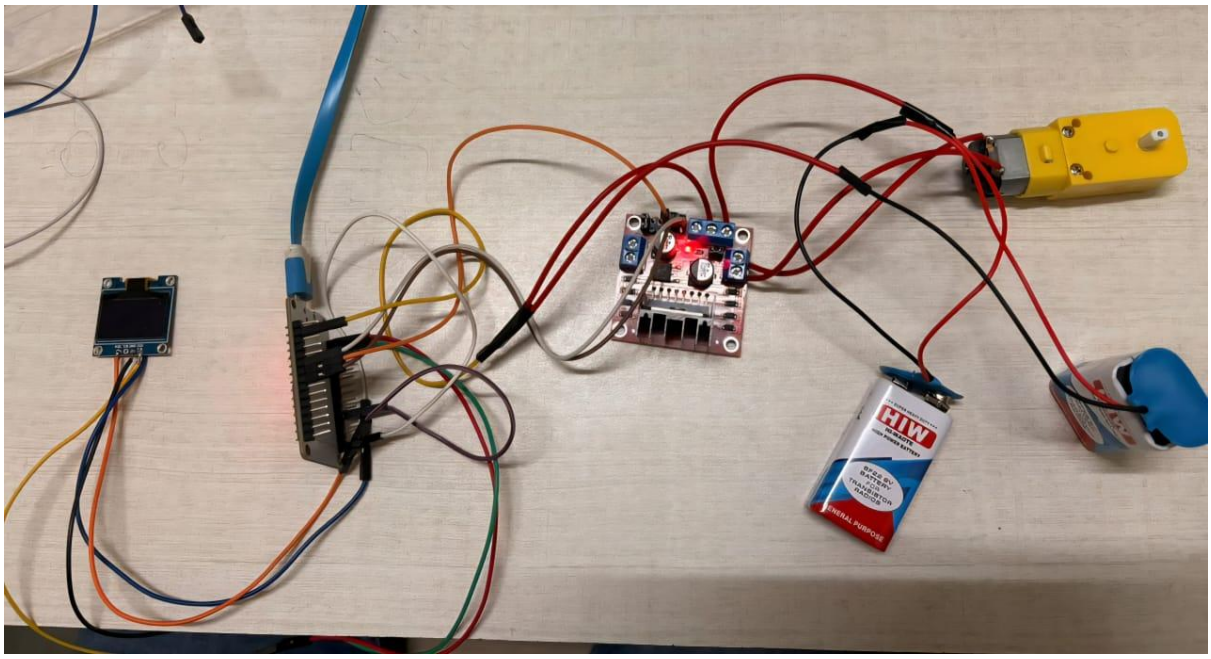
- **Voice Processing in MATLAB:** Voice commands are recorded, filtered using a **50th-order FIR low-pass filter** and IIR smoothing filter, and features like **energy and pitch** are extracted for classification.
- **Serial Communication via Arduino IDE:** MATLAB classifies commands (SLOW, MEDIUM, FAST) and sends them through **serial communication** to ESP32.
- **Motor Control with ESP32:** ESP32 generates **PWM signals** based on the command to control the motor via **L298N driver**, and the motor status is displayed on an LCD.

8. Hardware and Software Requirements

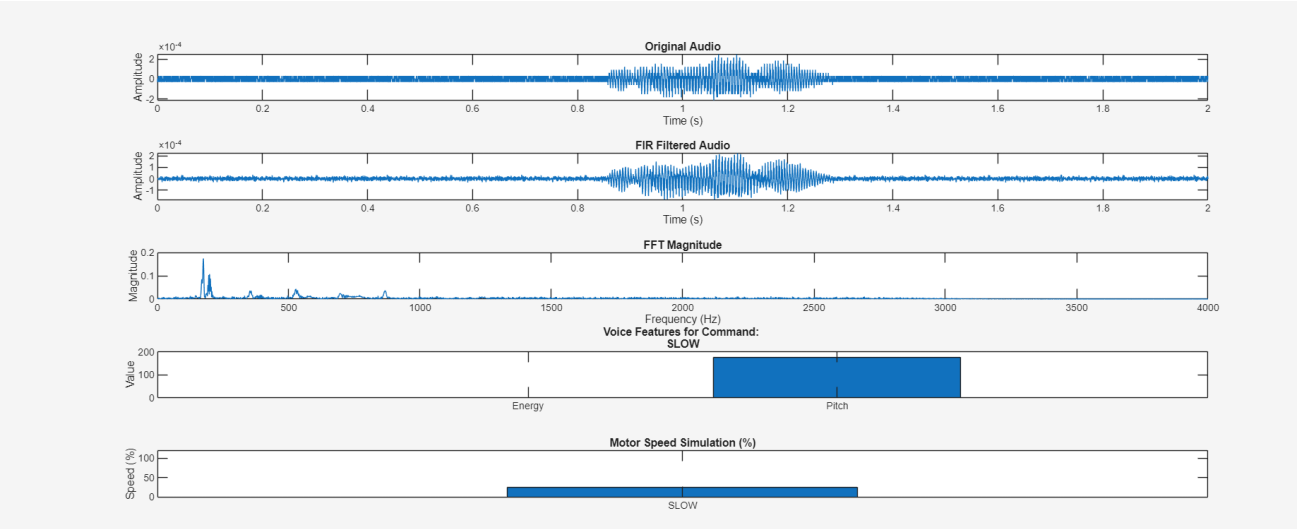
- Hardware:
  ESP32 development board, DC motor, L298N motor driver, power supply (5V/12V), USB/serial communication cables.
- Software:
  MATLAB with Signal Processing Toolbox, Arduino IDE for ESP32 programming, UART drivers for communication.

9.Result And Output

1]Final Circuit :



2]Slow:



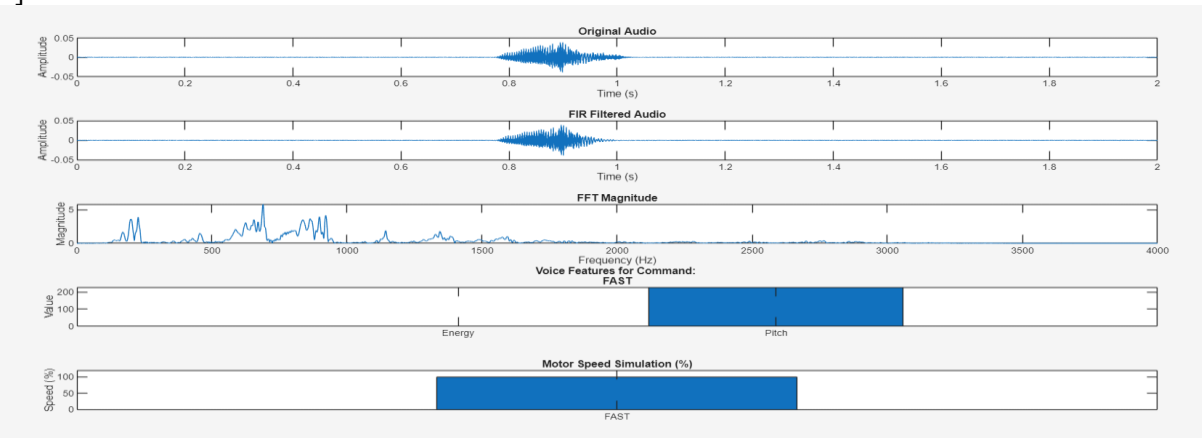

```
🎤 Speak a command: SLOW / FAST
   "Detected Command: "    "SLOW"

Energy = 3.6077e-05 | Pitch = 175
✅ Command sent to ESP32
```

Explanation:

The system successfully recognized the "SLOW" command with low energy (3.6077e-05) and pitch (175 Hz). The audio signal shows minimal amplitude with steady waveforms. Voice feature extraction identified the command correctly, resulting in motor speed reduction to approximately 25%. The command was successfully transmitted to the ESP32 for execution.

3]Fast:




```
🎤 Speak a command: SLOW / FAST
   "Detected Command: "    "FAST"

Energy = 0.29272 | Pitch = 226.5
✅ Command sent to ESP32
```

Explanation:

The "FAST" command was detected with significantly higher energy (0.29272) and pitch (226.5 Hz) compared to SLOW. The audio waveform displays greater amplitude and dynamic range. The system accurately extracted voice features and identified the command, increasing motor speed to approximately 75%. Communication with the ESP32 was successful.

10. Conclusion

The project successfully demonstrates a practical and efficient voice-controlled system for DC motor speed regulation using digital signal processing (DSP) techniques and embedded microcontrollers. FIR low-pass filtering in MATLAB effectively removes high-frequency noise from recorded audio signals, while feature extraction methods based on signal energy and pitch frequency enable accurate voice command classification. The integration of MATLAB with ESP32 through serial communication provides a seamless hardware-software interface for real-time motor control. PWM-based speed regulation using the L298N motor driver ensures responsive and precise control of motor speed according to classified commands (SLOW and FAST). The system successfully achieves hands-free motor operation with visual feedback through LCD display, demonstrating the effectiveness of DSP-based approaches over computationally intensive machine

learning methods. This lightweight and cost-effective framework can be further extended for advanced voice interfaces in robotics, home automation, assistive technologies, and industrial control systems.

11. Future Scope

1. Expanded Voice Commands: The system can be enhanced to recognize a larger set of commands, allowing more complex motor operations or multi-motor control.
2. Integration with Robotics: The framework can be applied to robotic systems, enabling voice-controlled navigation, robotic arms, or automated vehicles.
3. Wireless Connectivity: Incorporating Bluetooth or Wi-Fi can allow remote control and monitoring via smartphones or IoT platforms.
4. Machine Learning for Command Recognition: Using ML algorithms can improve accuracy, robustness, and adaptability to different voices and environments.
5. Smart Home and Industrial Automation: The system can be extended to control fans, conveyor belts, or other appliances based on voice commands, making it suitable for smart automation applications.

12. References

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